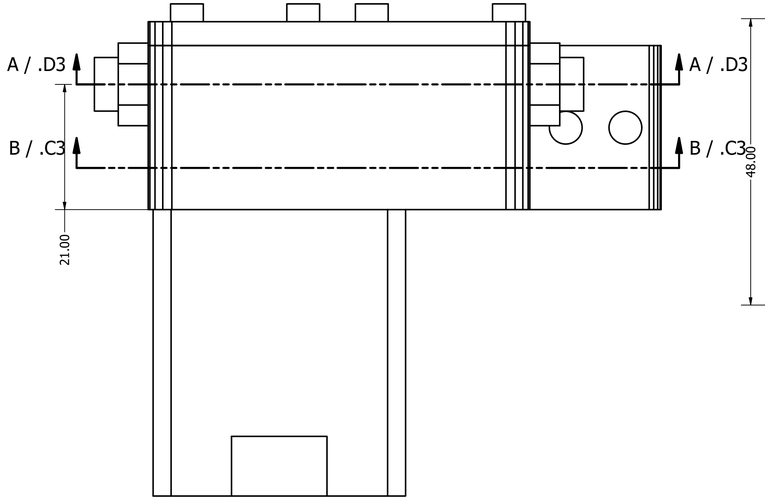
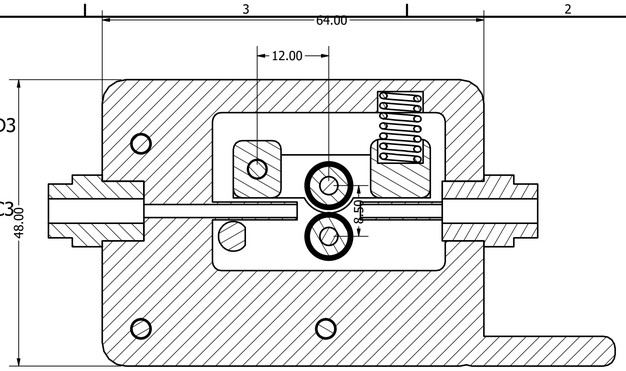
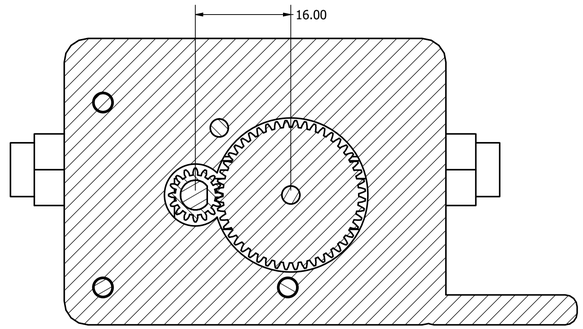
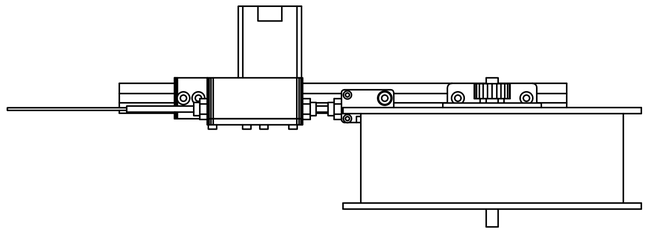


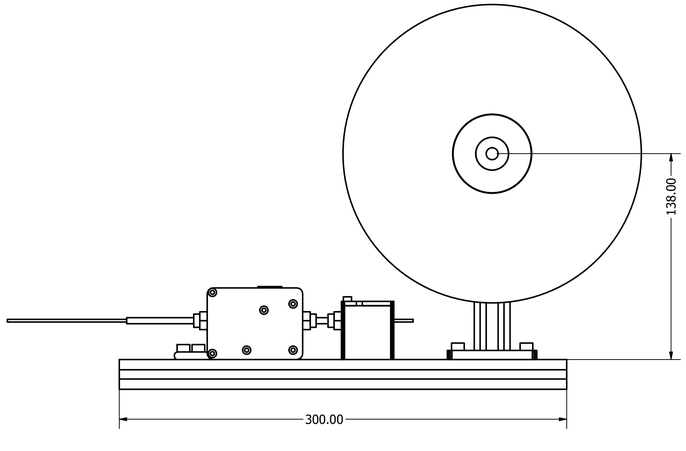


SECTION A-A
SCALE 2 : 1 / .D5



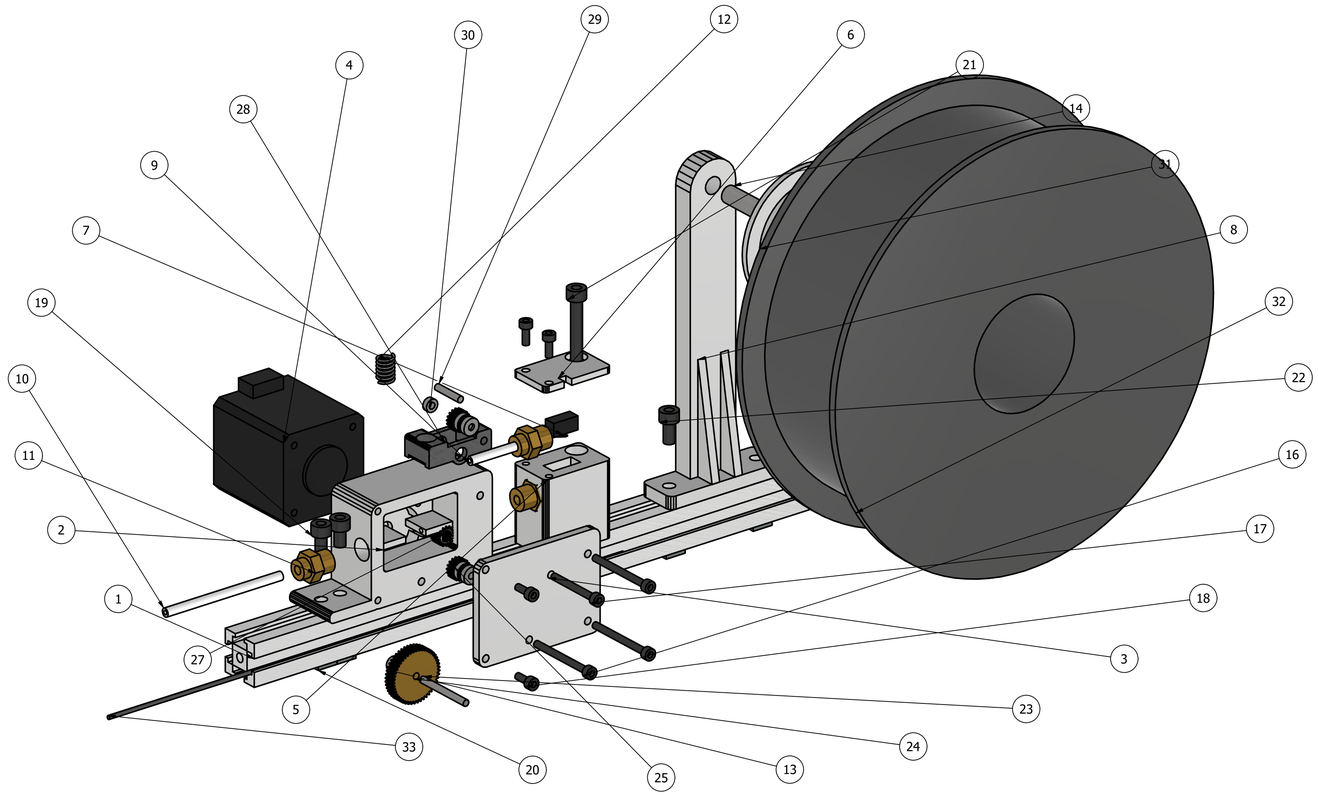
SECTION B-B
SCALE 2 : 1 / .D5





FILAMENTFEEDER - GEARED DUAL-DRIVE BOWDEN FEEDER, 1.75 MM FILAMENT
(DESIGNED AND CALCULATED, NOT BUILT OR PRINTED)
NEMA 17 STEPPERONLINE 17HS19-2004S1 -> 16T / 48T M0.5 (3:1) -> 2 x 17T DUAL-DRIVE GEARS, D EFF 7.36
415.2 STEPS/MM AT 1/16 (2.41 UM PER MICROSTEP)
IDLER: SPRING 8 x 1 x 18 (k 4.45 N/MM) ON A 2.0:1 LEVER: CLAMP 53 N
PUSH CAPACITY 64 N (MU 0.6 ASSUMED): SLIP ABOVE ~26 MM3/S (ASSUMED PLA MODEL)
PRINTED: BODY, COVER, RUNOUT BLOCK + LID, SPOOL UPRIGHT + HUB IN PETG; IDLER ARM IN PA6-CF
ASSEMBLY PARAMETERS Feed (MM) AND Lever (DEG); SHOWN AT 0 / 0
DIMENSIONS: DESIGN VALUES, EACH CHECKED AGAINST THE MODEL

DRAWN Jay Patel	11/2/2026			
CHECKED		TITLE FilamentFeeder - general assembly		
QA				
MFG				
APPROVED				
		SIZE D	DWG NO FF-DWG-001	REV
		SCALE 1 / 2	SHEET 1 OF 2	



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	FF-601	2020 V-slot / T-slot aluminium extrusion, 300 mm (6 mm slot, simplified profile)
2	1	FF-500	Feeder body, printed PETG: gear + drive pockets, 683ZZ seat, filament guides, 2 x PC4-M10 threads, motor face, mounting flange (2 x M5 to the 2020 rail); print back face down, supports in the gear pocket
3	1	FF-502	Feeder cover, printed PETG 4 mm: front 683ZZ seat, screw clearances
4	1	FF-101	Stepper motor StepperOnline 17HS19-2004S1 (NEMA 17, 59 Ncm, 2.0 A, 48 mm); 1.8 deg, 0.59 N m holding, 5 mm D shaft (envelope from the datasheet)
5	1	FF-504	Runout sensor block, printed PETG: filament channel, D2F switch pocket, PC4-M10 outlet, 1 x M5 to the rail
6	1	FF-505	Runout lid, printed PETG 3 mm: clamps the switch, wire exit
7	1	FF-311	Microswitch Omron D2F-01L (lever), 0.1 A, hinge lever (envelope from the datasheet): the filament holds the lever down
8	1	FF-506	Spool upright, printed PETG: 24 x 10 post with gussets, 60 x 20 foot (2 x M5 to the rail), 8 mm axle press bore
9	1	FF-309	PTFE tube 4 OD x 2 ID, 40 long (cut from a 1 m Capricorn XS / generic tube)
10	1	FF-310	PTFE tube 4 OD x 2 ID, 61 long (cut from a 1 m Capricorn XS / generic tube)
11	3	FF-308	Push-fit coupling PC4-M10 for 4 x 2 PTFE (envelope)
12	1	FF-307	Compression spring, music wire: 8 OD x 1.0 wire x 18 free, 6.5 active coils, k 4.45 N/mm (calculated); installed 12 = 26.7 N
13	2	FF-303	Deep groove ball bearing 683ZZ 3x7x3, shielded
14	1	FF-306	Spool axle, 8 mm steel rod x 100 (h9), pressed into the upright
15	2	FF-305	Deep groove ball bearing 608ZZ 8x22x7, shielded
16	3	FF-701	Socket head cap screw ISO 4762 M3 x 35 class 12.9 (motor, through cover + body)
17	1	FF-702	Socket head cap screw ISO 4762 M3 x 28 class 12.9 (idler pivot, into a heat-set insert)
18	4	FF-703	Socket head cap screw ISO 4762 M3 x 8 (cover, runout lid; heat-set inserts)
19	2	FF-707	Socket head cap screw ISO 4762 M5 x 8 (feeder flange into T-nuts)
20	5	FF-706	Drop-in T-nut M5 for 2020 6 mm slot
21	1	FF-705	Socket head cap screw ISO 4762 M5 x 30 (runout block into a T-nut)
22	2	FF-704	Socket head cap screw ISO 4762 M5 x 10 (feeder flange, upright foot into T-nuts)
23	1	FF-301	Main drive shaft, hardened steel 3 h6 x 30
24	1	FF-202	Spur gear 48T m0.5, brass, 5 wide, 3 bore with set screw (3:1 stage); brass, not POM: Lewis SF at the grip limit 1.3 in POM
25	2	FF-201	Dual-drive gear, hardened steel: 17T m0.5 spur teeth (the pair mesh) + hobbed section 8 OD with a 1.9 wide filament groove, effective grip diameter 7.36 (BMG pattern); 3 bore, 11 long. CAD is my envelope of the Bondtech-type gear: confirm the kit's dimensions
26	1	FF-102	Motor shaft (split out of the motor so it can turn in the model; part of FF-101)
27	1	FF-203	Motor pinion 16T m0.5, steel, 5 wide, 5 D bore
28	1	FF-503	Idler arm, printed PA6-CF: pivots on an M3 screw, carries the idler drive gear on 2 x MR63ZZ, spring pocket; lever ratio 2.00; PA6-CF, not PETG, because the spring load is constant (creep)
29	1	FF-302	Idler shaft, hardened steel 3 h6 x 16
30	2	FF-304	Deep groove ball bearing MR63ZZ 3x6x2.5, shielded
31	1	FF-501	Spool hub, printed PETG: 52.4 OD (3 mm wall) for 53-58 mm spool bores, 66 flange, two 608ZZ seats in 9 mm end walls
32	1	FF-401	1 kg filament spool, 200 OD x 68 wide, 53 bore (envelope; wound filament shown to r 88)
33	1	FF-402	1.75 mm filament (the moving part; part of the spool FF-401)

DRAWN Jay Patel	11/2/2026		
CHECKED		TITLE	FilamentFeeder - general assembly
QA			
MFG			
APPROVED		SIZE D	DWG NO FF-DWG-001 REV
		SCALE 0.80 : 1	SHEET 2 OF 2